

PUSHPAK PATIL

AWS Cloud Architect | DevOps Engineer | FinOps & AIOps Product Builder

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VISA SPONSORSHIP REQUIRED: Indian national seeking UK Skilled Worker visa sponsorship. 6+ years AWS cloud architecture & backend development expertise with 60% cost savings and 100% compliance in financial sector.

PROFESSIONAL SUMMARY

AWS Cloud Architect and DevOps Engineer with 6+ years of experience designing, developing, and implementing large-scale cloud infrastructure and backend services for multinational financial institutions. Expert in AWS architecture (EC2, VPC, EKS, RDS, Lambda, IoT Core), Python backend development, Infrastructure as Code (Terraform), and full software development lifecycle management. Proven expertise in IoT-enabled cloud services, real-time sensor data processing, ML model deployment, and time-series data pipelines. Track record: 60% cost reduction, 100% PCI DSS/ISO 27001 compliance, RTO under 10 minutes for DR solutions. Experienced in DevOps automation, CI/CD pipelines, and architecting scalable, highly available systems. Seeking UK opportunities with visa sponsorship.

TECHNICAL SKILLS

AWS Cloud Architecture: EC2, VPC, EKS, RDS, IAM, Lambda, S3, DynamoDB, IoT Core, IoT Analytics, Kinesis Data Streams, Control Tower, Organizations, Security Hub, GuardDuty, Config, multi-region architectures, high availability design

Backend Development: Python (Flask, FastAPI, Django), Node.js, API design and development, RESTful services, microservices architecture, serverless backend (Lambda), event-driven architectures

IoT & Real-Time Data: AWS IoT Core, IoT device provisioning and management, MQTT protocol, real-time sensor data ingestion, time-series data processing, streaming analytics, Kinesis Data Analytics, IoT-enabled cloud service infrastructure

Machine Learning & Data Pipelines: ML model deployment on AWS (SageMaker, Lambda), time-series forecasting, anomaly detection algorithms, data pipeline orchestration, ETL processes, real-time data processing for ML inference

Infrastructure as Code: Terraform, Terragrunt, AWS CloudFormation, Python automation, Shell scripting, Lambda State Machines, Step Functions, Policy-as-Code

DevOps & CI/CD: Jenkins, GitHub Actions, GitLab CI, automated deployment pipelines, blue/green & canary deployments, Docker containerization, Kubernetes/EKS, software development lifecycle management, agile practices

Monitoring & Observability: Prometheus, Grafana, CloudWatch, custom metrics, log aggregation, distributed tracing, automated alerting, AIOps platforms, ML-based anomaly detection

Security & Compliance: Cloud Security Posture Management (CSPM), IAM policy design, WAF & Shield, PCI DSS, CIS Benchmarks, ISO 27001 compliance, automated security auditing, zero-trust architecture

PROFESSIONAL EXPERIENCE

Cloud Solution Architect | Applied Cloud Computing, Thane, India

March 2025 – Present

- Architect and implement scalable AWS cloud infrastructure supporting IoT-enabled services, real-time sensor data ingestion, and ML model deployment for multinational financial technology clients
- Design and develop Python backend services using Flask and FastAPI, implementing RESTful APIs for IoT device management, time-series data processing, and ML inference endpoints
- Manage complete software development lifecycle from requirements gathering through deployment, implementing agile methodologies and DevOps best practices for continuous delivery
- Build and maintain data pipelines for real-time sensor data processing using AWS Kinesis, Lambda, and DynamoDB, enabling time-series analytics and ML-powered insights
- Lead DevOps automation initiatives: provisioning cloud infrastructure with Terraform, configuring CI/CD pipelines, implementing automated testing and deployment for backend services

Senior Cloud Engineer | Applied Cloud Computing, Mumbai, India

March 2022 – February 2025

- Designed and implemented AWS Landing Zones with Terraform for financial institutions (SBGIC, Zuno, HDBFS), managing complex multi-account environments with Control Tower and Organizations
- Developed Python automation scripts for cloud resource provisioning, cost optimization, security compliance monitoring, and operational workflow automation
- Implemented DevOps pipelines using Jenkins and GitLab CI for automated testing, deployment, and infrastructure updates across development, staging, and production environments
- Architected and deployed Kubernetes EKS clusters for microservices-based applications, implementing container orchestration, auto-scaling, and service mesh configurations
- Achieved 100% compliance (PCI DSS, CIS Benchmarks) through automated Config Rules, Security Hub monitoring, and custom compliance automation tools
- Delivered 20-60% cost reduction through intelligent resource optimization, automated rightsizing recommendations, and serverless architecture adoption

KEY PROJECTS & ACHIEVEMENTS

IoT-Enabled Banking Services Platform

- Architected scalable AWS IoT infrastructure for real-time ATM and branch sensor monitoring, processing 1M+ sensor readings per day using IoT Core, Kinesis, and Lambda
- Developed Python backend services for IoT device provisioning, data ingestion APIs, and time-series anomaly detection using ML models deployed on SageMaker
- **Result: Reduced ATM downtime by 45%, enabled predictive maintenance, achieved sub-second latency for real-time alerts**

Bandhan Bank – AIOps Platform with ML-Based Analytics

- Implemented comprehensive AIOps platform with ML-based anomaly detection for time-series metrics, automated incident response, and AI-powered root cause analysis
- Built Python data pipelines for metrics ingestion, preprocessing time-series data, training ML models, and deploying inference endpoints for real-time anomaly detection
- **Result: Reduced incident response from hours to minutes, achieved 100% audit readiness with zero critical security findings**

ATLAS – Automated Multi-Region DR with Software-Defined Orchestration

- Designed fully automated disaster recovery solution using Python Lambda State Machines, implementing intelligent health monitoring, multi-region failover logic, and automated rollback
- Developed backend orchestration layer for DR testing automation, integrating with CI/CD pipelines for continuous DR validation
- **Result: Zero-touch failover with RTO under 10 minutes, reduced DR testing from 8 hours to 30 minutes**

Edelweiss – Cloud-Native Microservices Platform

- Led migration of 50+ monolithic applications to cloud-native microservices on EKS, implementing DevOps practices, containerization, and automated deployment pipelines
- Established complete software development lifecycle with GitLab CI/CD, automated testing, blue/green deployments, and canary releases
- **Result: 35% cost reduction, improved deployment frequency from monthly to daily, enhanced scalability and reliability**

HDBFS – Compliance Automation & Infrastructure Optimization

- Built automated compliance monitoring infrastructure with Python scripts, Terraform automation, Config Rules, and Security Hub for PCI DSS/CIS alignment
- **Result: 100% compliance score, 60% infrastructure cost reduction, automated weekly compliance reporting saving 95% manual effort**

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate**
- **AWS Certified Security – Specialty**
- IBM Vault Automation Practitioner Advanced

EDUCATION

Master of Science in Organic Chemistry – Mumbai University (2015-2017)

Bachelor of Science in Mathematics & Computer Science – Mumbai University (2014-2015)

PRODUCTS BUILT

Independently designed, developed, and commercialised cloud-native SaaS products for the BFSI sector:

Nimbus FinOps AI — Multi-Tenant Cloud FinOps Platform for BFSI | finops.acc.ltd/dashboard | github.com/p05-cloud/zuno-finops

- Built a production-grade, multi-tenant Cloud FinOps SaaS platform for banking and financial services clients. Live at Poonawalla Fincorp (PFL), BHFL, and Axis Securities. Integrates AWS, Azure, GCP, and Kubernetes billing under a FOCUS-spec normalised data model with AI-powered cost recommendations, anomaly detection, rightsizing engine, Savings Plan optimisation, tagging governance, and budget alerting.
- **Tech Stack:** Next.js 15 (App Router), TypeScript, PostgreSQL 16, Redis 7, BullMQ, Prisma ORM, Zod, Tailwind CSS, Docker, Helm, GitHub Actions CI/CD. Monorepo (Turborepo), FOCUS schema normalisation, multi-tenant Row-Level Security in PostgreSQL.
- **Achievement:** Live in production at 3 BFSI clients. Reduced manual cost review effort by 80%, automated compliance governance across 4+ cloud accounts per client.

InfraOps AI — AIOps Platform for Automated L1 Incident Remediation |
github.com/p05-cloud/infraops-ai

- Building an AI-powered AIOps platform to eliminate L1 operational errors through intelligent monitoring, pattern recognition, and automated remediation. Uses ML models to detect infrastructure anomalies, classify incident severity, and trigger auto-remediation via AWS Systems Manager and Lambda — reducing on-call burden and MTTR for cloud operations teams.
- **Tech Stack:** AWS CloudWatch, Lambda, Systems Manager, Prometheus, Grafana, Python (ML inference), event-driven architecture.
- **Target:** 70% reduction in L1 ticket volume and sub-5-minute automated remediation for common cloud infrastructure failures.

AWARDS & RECOGNITION

- Excellence in Cloud Cost Management Award
- Cross-Team Collaboration Excellence Award
- Anchor Project Excellence Award
- Led Disaster Recovery drills and restoration testing across banking client accounts
- Created 18 cloud infrastructure automations for process optimization and time savings

References available upon request | Open to relocation within the UK with visa sponsorship